LESSON 10.9
WRITING TWO-STEP EQUATIONS FROM REAL-WORLD CONTEXTS

LESSON INTRODUCTION

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical context or real-world context, where all terms are rational numbers.

LEARNING TARGETS

- I can write two-step equations to represent real-world contexts.
- I can solve two-step equations within a real-world context.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

WORKING TOWARD

MA.8.AR.2.1 Solve multi-step linear equations in one variable with rational number coefficients. Include equations with variables on both sides.

ESSENTIAL QUESTIONS

- How do you write two-step equations to represent real-world contexts?
- How do you interpret the solution for a real-world context?

LESSON NARRATIVE

This is the second of three lessons in which students develop their skills in writing equations of the form $px + q = r$ and $p(x + q) = r$. Students explore constants and unknowns in real-world contexts. Students extend this to write and solve two-step equations from real-world contexts.

COMPONENT SUMMARY

10.9.1 WARM-UP *(~5 MIN.)*

Identify and correct errors in sample of student work

10.9.2 GUIDED INSTRUCTION *(15 MIN.)*

Write and solve two-step equations to represent real-world contexts

10.9.3 TRY IT *(10 MIN.)*

Practice writing and solving two-step equations from real-world contexts

10.9.4 GUIDED PRACTICE *(10 MIN.)*

Write and solve two-step equations from real-world contexts involving percentages

10.9.5 WRAP-UP *(~5 MIN.)*

Identify and correct errors in sample of student work

RESOURCES

GLOSSARY

Key Terms:

- Variables

Additional Terms:

- Commission
- Discount

MATERIALS

- Calculators

ADDITIONAL PREPARATION

- Consider creating and displaying an anchor chart with the steps needed to write and solve two-step equations from real-world problems.
- Consider a context in which your students can relate to explain that p in px + q = r is often a rate and that the q is often a beginning value. Use a simple example such as pouring water into a cup that already contains water. The original height is q and the rate at which the water is being poured in is p.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may make errors in their placement of decimals when working with money and percentages.
- Students may struggle defining the unknown.
- Students may struggle understanding the p in px + q = r is often a rate and that the q is often a beginning value.
- Students may struggle understanding the p in p(x + q) = r is often the number of groups.

THINGS TO CONSIDER

- Review how to convert percentages to decimals and remind students to multiply the decimal representation of the percentage by the quantity to determine the percentage of the quantity.
- If time is an issue, consider assigning Try It for homework.
- Allowing students to use calculators during the lesson will help students focus on reading and interpreting the real-world contexts.

LITERACY AND LANGUAGE ROUTINES

Take Turns

In 10.9.2 Guided Instruction, students discuss the known, unknown, variable, and constant quantities from a real-world problem. Then, they collaboratively write and solve the equation that represents the real-world problem. Ask students to take turns explaining, justifying, agreeing, or disagreeing, and asking clarifying questions as they complete the problems.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In both the Warm-Up and Wrap-Up, students identify and correct errors in samples of student work. These activities will help students check their own work and avoid making mistakes themselves in the future. Describing the errors will also help students develop their ability to construct viable arguments and critique the reasoning of others.

DIFFERENTIATE INSTRUCTION

Provide students with the flexibility to solve the two-step equations using their preferred method in 10.9.3 Try It and 10.9.4 Guided Practice. During the debrief, ask students to share their methods and reflect on whether a more efficient method exists.

10.9.1 WARM-UP: BELL WORK

TEACHER MOVES

Error Analysis activities will help students check their own work and avoid making mistakes themselves in the future. Describing the errors will also help students develop their ability to construct viable arguments and critique the reasoning of others.

10.9.1 Warm-Up: Bell Work

Tremaine solved the equation $-10 = -8.5 + 2x$ and checked his solution. His work and check are shown.

Work	Check
$\frac{-10}{2} = -8.5 + \frac{2x}{2}$	$-10 = -8.5 + 2(3.5)$
$-5 = -8.5 + x$	$-10 = -8.5 + 7$
$+8.5 \quad +8.5$	$-10 \neq -1.5$
$3.5 = x$	

Tremaine knows his solution is incorrect, but he is not sure where he went wrong.

1. Explain how Tremaine knows his solution is incorrect.
   Tremaine knows his solution is not correct because when he substituted it back into the original equation, it did not make both sides of the equation equivalent.

2. Describe the error he made.
   When Tremaine used the division property of equality, he did not divide every term by 2. He forgot to divide -8.5 by 2.

3. Find the correct solution to the equation $-10 = -8.5 + 2x$. Show your work.
   $-\frac{10}{2} = \frac{-8.5 + 2x}{2}$
   $-5 = -4.25 + x$
   $-5 + 4.25 = -4.25 + 4.25 + x$
   $-0.75 = x$

10.9.2 Guided Instruction: Writing Two-Step Equations from Real-World Contexts

When writing equations to represent a real-world context, it is helpful to start by considering the known and unknown quantities involved in the situation.

1. Amanda is draining her pool for resurfacing. The pool began with 12,340 gallons of water and it is draining at a rate of 780 gallons per hour. How long has the pool been draining if currently there are 5,125 gallons in the pool?

   a. Complete the table to describe the known and unknown quantities in this situation.

Known Quantities	Unknown Quantities
○ *Beginning amount of water in the pool.* ○ *How fast the water is draining.* ○ *The amount of water that is left in the pool.*	○ *How long the pool has been draining for.*

   b. With your partner, discuss how the quantities are related in this situation.
   Answers will vary.
   If we take the amount of water at the beginning and subtract the product of the amount drained per hour times the number of hours, you will get the remaining amount of water.

When writing equations to represent real-world contexts, **any variables used must be defined** to identify which quantity each variable in the equation represents.

   c. Define the unknown quantity using a variable.
   Answers will vary.
   t = the number of hours the pool has been draining.

   d. Write an equation that could be used to find the unknown quantity using the variable you defined.
   $12{,}340 - 780t = 5{,}125$

   e. Solve the equation.
   $12{,}340 - 780t = 5{,}125$
   $12{,}340 - 12{,}340 - 780t = 5{,}125 - 12{,}340$
   $-780t = -7215$
   $\frac{-780t}{-780} = \frac{-7215}{-780}$
   $t = 9.25$

   f. Explain what the solution means in this context.
   The pool has been draining for 9 ¼ hours or 9 hours and 15 minutes.

2. Six friends met for a book club at a local coffee shop. Each person ordered a small cup of coffee and a muffin. The total for the order, including taxes, was \$37.56. If the cost of a small coffee is \$3.79 after taxes, find the cost of one muffin after taxes.

   a. Complete the table to describe the known and unknown quantities in this situation.

Known Quantities	Unknown Quantities
○ *The total number of friends.* ○ *The total cost of the order.* ○ *The cost of a cup of coffee.*	○ *The cost of one muffin.*

10.9.2 GUIDED INSTRUCTION: WRITING TWO-STEP EQUATIONS FROM REAL-WORLD CONTEXTS

TEACHER MOVES

The lesson is scaffolded to help students work through the process of setting up an equation for a real-world context so that this process is internalized. Consider refraining from providing a set of steps that students should follow. Allow the student to utilize the scaffolding to aid in their own thinking process. Allow students the ability to think flexibly about the process, thus solidifying their understanding in a way that works for them. If a student seeks further structure, then a step-by-step guide may be helpful for that student. Also, consider that students may write an equivalent equation. Consider how to allow for students to think flexibly. It may be helpful to state the equations should be in the form px + q = r or p(x + q) = r.

TEACHER MOVES

Students may write the equation 12,340-780=5,125 without considering the hours the pool has been draining. Ask students if that equation considers the amount of time the pool has been draining. Then, ask them to consider which value will change with time. Relate this to the unknown quantity in question 1a.

TEACHER MOVES

Allow the students to answer question 2a. Then, have students share their unknown quantities. If necessary, ask your students these questions:

- How much does a muffin cost?
- How much does a small coffee cost?
- How many people ordered a cup of coffee and a muffin?
- How many coffees and muffins were purchased?
- What was the total cost of the order?

10.9.2 GUIDED INSTRUCTION: WRITING TWO-STEP EQUATIONS FROM REAL-WORLD CONTEXTS (CONTINUED)

TEACHER MOVES

Help students understand why the value of 6 is being distributed to (3.79+m). Relating it to six groups of muffins and coffees may help students understand. Also, some students may distribute in their head and write the equation 6m+22.74=37.56. Both equations should be acceptable and continue to help student understanding of equivalent equations. Other students may write the equation as m+3.79=6.26. These students are showing an equation for one part of the group. Allowing for student flexibility is important.

DIFFERENTIATE INSTRUCTION

Consider having students draw a diagram or other visual representation to represent the 6 groups of muffins and coffees.

b. Define the unknown quantity using a variable.
Answers will vary.
$m =$ *the cost of one muffin after tax.*

c. Write an equation that could be used to find the unknown quantity using the variable defined.
$6(3.79 + m) = 37.56$

d. Solve the equation.
$6(3.79 + m) = 37.56$
$\frac{6(3.79 + m)}{6} = \frac{37.56}{6}$
$3.79 + m = 6.26$
$3.79 - 3.79 + m = 6.26 - 3.79$
$m = 2.47$

e. Explain what the solution means in this context.
One muffin costs $2.47 *after tax.*

10.9.3 Try It: Writing Two-Step Equations from Real-World Contexts

1. The average toddler's height increases by 0.2 inches per month. Maurice's youngest brother James, who is a toddler, is 35.45 inches tall. If he grows at the average rate, how long will it take James to reach a height of 37 inches?

   a. Define the unknown quantity using a variable.
   Answers will vary.
   $m =$ *the number of months it will take James to reach a height of* 37 in.

   b. Write an equation that could be used to find the unknown quantity using the variable defined.
   $35.45 + 0.2m = 37$

   c. Solve the equation.
   $35.45 + 0.2m = 37$
   $35.45 - 35.45 + 0.2m = 37 - 35.45$
   $.2m = 1.55$
   $\frac{0.2m}{.2} = \frac{1.55}{0.2}$
   $m = 7.75$

   d. Explain what the solution means in this context.
   It will take James 7.75 *months to reach a height of* 37 *inches.*

10.9.3 TRY IT: WRITING TWO-STEP EQUATIONS FROM REAL-WORLD CONTEXTS

DIFFERENTIATE INSTRUCTION

Consider allowing partners to discuss their reasoning and check each other's work. This partner work will provide auditory and visual entry-points as students hear and see different methods of solving the equation.

10.9.4 GUIDED PRACTICE: WRITING TWO-STEP EQUATIONS FROM REAL-WORLD CONTEXTS INVOLVING PERCENTAGES

TEACHER MOVES

Review the meanings of the terms “commission” and “discount.” Remind students that commission represents a percentage increase and discount represents a percentage decrease. Review how to convert a percentage to a decimal..

QUESTIONING

Consider sharing sales fliers with students and asking them to write and solve equations involving percentages.

10.9.4 Guided Practice: Writing Two-Step Equations from Real-World Contexts Involving Percentages

1. Victoria works as a salesperson. She earns $\$350$ per week, plus 10% commission on her weekly sales. If Victoria needs to earn $\$800$ this week, determine how much she needs to sell this week.

   a. Define the unknown quantity using a variable.
   Answers will vary.
   s = the amount that Victoria still needs to sell this week.

   b. Write an equation that could be used to find the unknown quantity using the variable defined.
   $350 + 0.1s = 800$

   c. Solve the equation.
   $350 + 0.1s = 800$
   $350 - 350 + 0.1s = 800 - 350$
   $0.1s = 450$
   $\frac{0.1s}{0.1} = \frac{450}{0.1}$
   $s = 4{,}500$

   d. Explain what the solution means in this context.
   Victoria still needs to sell $\$4{,}500$ worth of product to earn a commission of $\$800$ this week.

2. Maddox has a coupon for $\$10$ off his purchase at Foot Zone. He found a pair of shoes he wants that is marked 20% off. If Maddox paid $\$36.20$ before taxes were applied, what was the original price of the shoes?

   a. Define the unknown quantity using a variable.
   Answers will vary.
   p = the original price of the shoes before taxes.

   b. What percentage of the original price of the shoes did Maddox pay?
   80% of the original price.

   c. Write an equation that can be used to find the unknown quantity using the variable defined.
   $0.80p - 10 = 36.20$

   d. Solve the equation.
   $0.80p - 10 = 36.20$
   $0.80p - 10 + 10 = 36.20 + 10$
   $0.80p = 46.20$
   $\frac{0.80p}{0.80} = \frac{46.20}{0.80}$
   $p = 57.75$

   e. Explain what the solution means in this context.
   The original price of the shoes was $\$57.75$.

10.9.5 Wrap-Up: Error Analysis

Mr. Humphries told his class he has some quarters and four dimes in his pocket, worth a total of $\$2.40$.

Devin wrote the equation $25x + 40 = 2.40$ to represent the situation.

1. Mr. Humphries told Devin his equation did not correctly represent the situation. Write an equation that correctly represents the situation.
   Answers will vary.
   $0.25x + 0.40 = 2.40$ *or* $25x + 40 = 240$

2. If x represents the number of quarters, solve the equation you created in Question 1 to determine how many quarters Mr. Humphries has.
   Answers will vary.
   $25x + 40 = 240$
   $25x + 40 - 40 = 240 - 40$
   $\frac{25x}{25} = \frac{200}{25}$
   $x = 8$
   Mr. Humphries has 8 quarters.

10.9.5 WRAP-UP: ERROR ANALYSIS

QUESTIONING

For students who finish early, encourage them to find all of Devin's errors in his equation rather than just one.